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MTH 240 Midterm 1

Q1 10

10 Marks 1. Evaluate the integral

$$\int_1^e \frac{\ln x}{x^2}\,dx$$

$u = \ln x \qquad du = \frac{1}{x}\,dx$

$dv = \frac{1}{x^2}\,dx \qquad v = -\frac{1}{x}$

$$\int_1^e \frac{\ln x}{x^2}\,dx = \left[-\frac{1}{x}\ln x\right]\Big|_1^e - \int_1^e -\frac{1}{x^2}\,dx$$

$$= \left[-\frac{1}{x}\ln x\right]\Big|_1^e + \int_1^e x^{-2}\,dx$$

$$= \left[-\frac{1}{x}\ln x - \frac{1}{x}\right]\Big|_1^e$$

$$= \left(-\frac{1}{e}\ln e - \frac{1}{e}\right) - \left(-\ln 1 - \frac{1}{1}\right)$$

$$= \left(-\frac{1}{e} - \frac{1}{e}\right) - (-1)$$

$$= -\frac{2}{e} + 1$$

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Q2 5

10 Marks

2. Evaluate the integral

$$\int_0^{\pi/3} \frac{\sin^3 x}{\cos^4 x} + \sin(2x)\cos^3 x \, dx$$

$$\int_0^{\pi/3} \frac{(1-\cos^2 x)\sin x}{\cos^4 x}\,dx + \int_0^{\pi/3} (2\sin x\cos x)\cos^3 x\,dx$$

Sub $u = \cos x$

$du = -\sin x\,dx$

$-du = \sin x\,dx$

(Same substitution)

$x = \frac{\pi}{3} \Rightarrow u = \frac{1}{2}$

$x = 0 \Rightarrow u = 1$

$$\int_1^{1/2} \frac{1-u^2}{u^4}\,du + \int_1^{1/2} 2u^4\,du$$

$$= \int_1^{1/2} \frac{1}{u^4} - \frac{1}{u^2}\,du + \left[\frac{2}{5}u^5\right]\Big|_1^{1/2}$$

$$= \left[-\frac{1}{3u^3} + \frac{1}{u} + \frac{2}{5}u^5\right]\Big|_1^{1/2}$$

$$= \left[\left(-\frac{1}{3\left(\frac{1}{2}\right)^3} + 2 + \frac{2}{5}\left(\frac{1}{2}\right)^5\right) - \left(-\frac{1}{3} + 1 + \frac{2}{5}\right)\right]$$

$$= \left(-\frac{2^3}{3} + 2 + \frac{1}{5(2)^4}\right) - \left(\frac{-5+15+6}{15}\right)$$

$$= \left(-\frac{8}{3} + 2 + \frac{1}{80}\right) - \frac{16}{15}$$

$$= \frac{-8+6}{3} + \frac{1}{80} - \frac{16}{15}$$

$$= \frac{-2}{3} + \frac{1}{80} - \frac{16}{15}$$

$$= \frac{-160+80}{240} - \frac{16}{15}$$

$$= -\frac{80}{240} - \frac{16}{15}$$

$$= -\frac{1}{3} - \frac{16}{15}$$

$$= -\left(\frac{5+16}{15}\right)$$

$$\boxed{= -\frac{21}{15}}$$

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10 Marks 3. Evaluate the integral

Q3 9

$$\int \frac{x}{(3+2x-x^2)^{3/2}}\,dx$$

$$-x^2+2x+3 = -(x^2-2x)+3 = -(x^2-2x+1)+4 = -(x-1)^2+4$$

$$\int \frac{x}{(4-(x-1)^2)^{3/2}}\,dx$$

Sub $u = x-1$
$du = dx$
$x = u+1$

$$\int \frac{u+1}{(4-u^2)^{3/2}}\,du$$

Sub $u = 2\sin\theta$
$du = 2\cos\theta\, d\theta$ $\Rightarrow \sin\theta = \frac{u}{2} = \frac{x-1}{2}$

$$\int \frac{2\sin\theta+1}{(4-4\sin^2\theta)^{3/2}}\,2\cos\theta\, d\theta$$

$$= \int \frac{2\sin\theta+1}{\cancel{2}\cos^{3}\theta}\,\cancel{2}\cos\theta\, d\theta$$

-1

$$= \int \frac{2\sin\theta+1}{\cos^2\theta}\,d\theta$$

$$= \int \frac{2\sin\theta}{\cos^2\theta} + \frac{1}{\cos^2\theta}\,d\theta$$

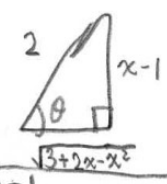
$$= 2\int \frac{\sin\theta}{\cos^2\theta}\,d\theta + \int \sec^2\theta\, d\theta$$

Sub $w = \cos\theta$
$dw = -\sin\theta\, d\theta$

$$= -2\int w^{-2}\,dw + \tan\theta + C$$

$$= -2\left(-\frac{1}{w}\right) + \tan\theta + C$$

$$= 2\sec\theta + \tan\theta + C$$



$$= 2\left(\frac{2}{\sqrt{3+2x-x^2}}\right) + \frac{x-1}{\sqrt{3+2x-x^2}} + C$$

$$\therefore \int \frac{x}{(3+2x-x^2)^{3/2}}\,dx = \frac{3+x}{\sqrt{3+2x-x^2}} + C$$

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Q4 10

10 Marks 4. Evaluate the integral

$$\int \frac{2x + 2x\sqrt[3]{x^2+5}}{3(x^2+5)^2 + 3(x^2+5)^{4/3}}\,dx$$

Hint: Consider a rationalizing substitution.

$u = \sqrt[3]{x^2+5}$

$x = \sqrt{u^3-5}$

$dx = \frac{3u^2}{2\sqrt{u^3-5}}\,du$

$$= \int \frac{2\sqrt{u^3-5} + 2\sqrt{u^3-5}\,(u)}{3u^6 + 3u^4}\left(\frac{3u^2}{2\sqrt{u^3-5}}\right)du$$

$$= \int \frac{\cancel{(2\sqrt{u^3-5})}(1+u)}{\cancel{(3u^2)}(u^4+u^2)}\left(\frac{\cancel{3u^2}}{\cancel{2\sqrt{u^3-5}}}\right)du$$

$$= \int \frac{1+u}{u^4+u^2}\,du = *$$

Partial Fraction Decomposition:

$$\frac{1+u}{u^2(u^2+1)} = \frac{A}{u} + \frac{B}{u^2} + \frac{Cu+D}{u^2+1}$$

$$1+u = Au(u^2+1) + B(u^2+1) + (Cu+D)u^2$$

Sub $u=0$: $\boxed{1 = B}$

Sub $u=1$: $2 = 2A+2+C+D$, $0 = 2A+C+D$ (1)

Sub $u=-1$: $0 = -2A+2-C+D$, $-2 = -2A-C+D$ (2)

Sub $u=2$: $3 = 10A+5+8C+4D$, $-2 = 10A+8C+4D$, $-1 = 5A+4C+2D$ (3)

(1) + (2): $0 = 2A+C+D$, $-2 = -2A-C+D$; $-2 = 2D$, $\boxed{D=-1}$

$4\times(1) - (3)$: $0 = 8A+4C-4$, $-1 = 5A+4C-2$; $1 = 3A-2$, $\boxed{A=1}$

Sub all into (1): $0 = 2(1)+C-1$, $\boxed{C=-1}$

$$* = \int \frac{1}{u} + \frac{1}{u^2} - \frac{u+1}{u^2+1}\,du$$

$$= \ln|u| - \frac{1}{u} - \int\frac{u}{u^2+1} - \int\frac{1}{u^2+1}\,du$$

$$= \ln|u| - \frac{1}{u} - \frac{1}{2}\ln|u^2+1| - \tan^{-1}u + C$$

$$= \ln\left|\sqrt[3]{x^2+5}\right| - \frac{1}{\sqrt[3]{x^2-5}} - \frac{1}{2}\ln\left((x^2+5)^{2/3}+1\right) - \tan^{-1}\left(\sqrt[3]{x^2+5}\right) + C$$

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Q5 10

10 Marks

5. Determine whether the following integrals are convergent or divergent. If they are convergent, find their value(s).

(a) $\displaystyle\int_2^{\infty} \frac{1}{x} - \frac{1}{x+1}\, dx$

$$\lim_{t\to\infty}\left[\int_2^t \frac{1}{x} - \frac{1}{x+1}\, dx\right]$$

$$= \lim_{t\to\infty}\Big[\ln|x| - \ln|x+1|\Big]_2^t$$

$$= \lim_{t\to\infty}\left[\ln\left(\frac{x}{x+1}\right)\right]\Big|_2^t$$

$$= \lim_{t\to\infty}\left[\ln\left(\frac{t}{t+1}\right)\right] - \ln\left(\frac{2}{3}\right)$$

$$= \ln\left(\lim_{t\to\infty}\frac{t}{t+1}\right) - \ln\left(\frac{2}{3}\right)$$

(L'Hopital's Rule)

$$= \ln\left(\lim_{t\to\infty} 1\right) - \ln\left(\frac{2}{3}\right)$$

$$= 0 - \ln\frac{2}{3} = \underline{\ln\left(\frac{3}{2}\right)}$$

$\therefore$ This integral is convergent and has a value of $\ln\frac{3}{2}$. 5

(b) $\displaystyle\int_0^{e} \ln x\, dx$

$$= \lim_{t\to 0^+}\int_t^e \ln x\, dx$$

$u = \ln x \quad du = \frac{1}{x}dx$

$dv = dx \quad v = x$

$$= \lim_{t\to 0^+}\left[(x\ln x)\Big|_t^e - \int_t^e dx\right]$$

$$= \lim_{t\to 0^+}\left[(x\ln x - x)\Big|_t^e\right]$$

$$= \lim_{t\to 0^+}\left[(e\ln e - e) - (t\ln t - t)\right]$$

$$= e - e - \lim_{t\to 0^+}(t\ln t - t)$$

$$= -\lim_{t\to 0^+}\left(\frac{\ln t}{t^{-1}}\right) + 0 \overset{\text{(L'hopitals Rule)}}{=} -\lim_{t\to 0^+}\frac{\left(\frac{1}{t}\right)}{-\frac{1}{t^2}} = \lim_{t\to 0^+} t = \underline{0}$$

5 . This integral converges and has a value of 0.

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Space1

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Space2